

Euclidean Geometry In Mathematical Olympiads

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Lemma 1.42: Common Centers



Week 1

geometry

incenter

Chapter 1

fgm

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collinear. Now if we can show that $AX \perp YZ$, $YB \perp ZX$, $ZC \perp XY$, it means that AX, YB, ZC are the angle bisectors of $\triangle ABC$ and altitudes of $\triangle XYZ$ which is what we want.

Archimedes15

1490 posts

Jun 8, 2020, 2:16 PM • 1

#11

Just let half of angle A be alpha, half of angle B be beta, half of angle C be gamma and then just use the fact that angles that subtend the same arcs are equal and then the problem gets destroyed bc angle ZYX is gamma + alpha and angle AXY is beta so AX is perpendicular to ZY and you can do the same for the rest of the altitudes.

Ilikeapos

1534 posts

Jun 20, 2020, 6:10 PM • 1

#12

My Solution

Notice that it suffices to prove that $YZ \perp XA$. The other altitudes follow similarly.

To do this, notice that $\angle AXZ = \frac{\angle C}{2}$ and $\angle YZX = \frac{\angle A + \angle B}{2}$. Adding these up gives 90° , so $YZ \perp XA$. Thus, applying the same logic, we have that the three lines are perpendicular and I is thus the orthocenter of XYZ .

Understanding...

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#13

Can someone check this please(because no one have mentioned it yet) ?

let $D = AX \cap YZ$

$A - I - X$ is collinear because of Incentre\excentre lemma

now

$\angle ZAX = \angle XAY = \angle BYX$ hence arc $ZX = \text{arc } YX$ which implies arc $ZA = \text{arc } AY$

so chord $AZ = AY$ and $\triangle AZD$ congruent to $\triangle AYD$, hence $YZ \perp XA$.
Done.

anything wrong ?

*This post has been edited 7 times. Last edited by Understandingmathematics, Jul 29, 2020, 8:14 AM
Reason: latex*

Understanding...

46 posts

Jul 29, 2020, 8:38 AM

#14

Bump \ Can someone pls check above..

squareman

Aug 2, 2020, 10:49 AM

#15



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